

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (IT) Examination

May 2013

IT308 COMPUTER NETWORKING-II

Date: 02.05.2013, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the hidden station and exposed station problem in wireless LAN. [05]
(b) Define the following term: [02]
1) SONET 2) Virtual Path
- Q - 2 (a) Explain the connection-oriented service of packet switching at network layer. [06]
(b) Draw and explain the architecture of IEEE 802.11. [04]
(c) Explain the PPP protocol in Point-to-Point WAN. [04]

OR

- Q - 2 (a) List and explain the different network layer issues. [06]
(b) Explain the network layer services provided at each router in detail. [04]
(c) Explain the advantages of frame relay over X.25 protocol in switched WAN. [04]
- Q - 3 (a) Justify the following statements with reason: [06]
1) The request and reply message can be used to bind a physical address to an IP address in a SVC situation.
2) The inverse request and inverse reply message can be used to build the server's mapping table.
- (b) Consider the following information: [04]
Original timestamp: 50 Receive timestamp: 60
Transmit timestamp: 61 Return timestamp: 67
What is the round-trip time? What is the difference between sender clock and receiver clock?
- (c) What is the need of proxy ARP? How it works? [04]

OR

- Q - 3 (a) Draw and explain in detail the different types of IGMP message format. [06]
(b) Calculate the Checksum in ICMP for following information: [04]

8	0	0
1		9
TEST		

- (c) What is Logical IP Subnet (LIS) in ATMARP? [04]

SECTION – II

Q - 4 (a) Attempt Following.

[07]

- 1) What is the network address if one of the addresses is 167.199.170.82/27?
- 2) What is the maximum size of IP header?
- 3) Using 7-bit sequence numbers, what is the maximum size of sender and receiver Windows for Go-back-N?
- 4) Show the original (unabbreviated) form of following IPv6 address:
0 : 0 : FFFF ::
- 5) What is pulling in transport layer?
- 6) Define socket address.
- 7) What is loop back address?

Q - 5 (a) Draw and explain connection oriented service of transport layer.

[06]

(b) Differentiate following:

[04]

- 1) Classful and Classless IPv4 addressing.
- 2) TCP and UDP

(c) What are the different types of delivery for IP packet? Explain each one. A host with IP address 137.23.56.23/16 sends a packet to a host with IP address 137.23.67.9/16. Which type of delivery?

[04]

OR

Q - 5 (a) An organization is granted the block 130.34.12.64/26. The organization needs to have four sub networks. What are the subnet addresses and the range of addresses for each subnet?

[06]

(b) Explain stop and wait protocol of transport layer.

[04]

(c) Draw an IP header format.

[04]

Q - 6 (a) List out the transition strategies of IPv4 to IPv6 and explain each one in detail.

[06]

(b) What is fragmentation in IPv4? List out the fields related to fragmentation? Explain any one in brief.

[04]

(c) Explain different types of links in OSPF.

[04]

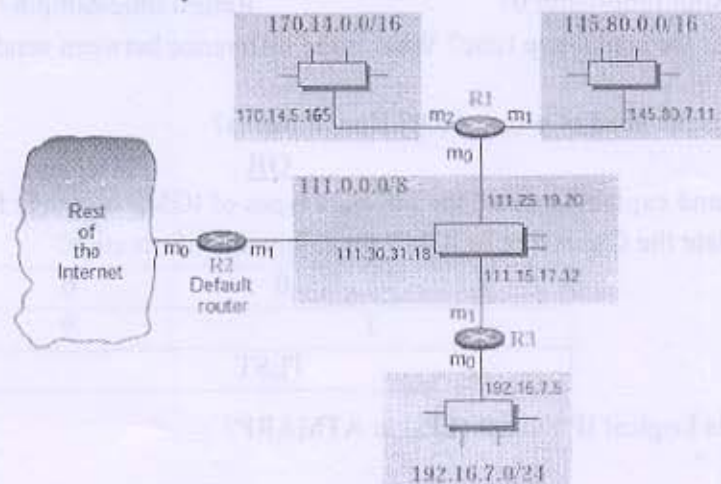
OR

Q - 6 (a) List out TCP timers and explain each one in detail.

[06]

(b) Figure shown is an imaginary part of the Internet. Show the routing tables for router R1. Router R1 receives a packet with destination address 192.16.7.14. Show how the packet is forwarded.

[04]



(c) Explain multicasting and multiple unicasting.

[04]
